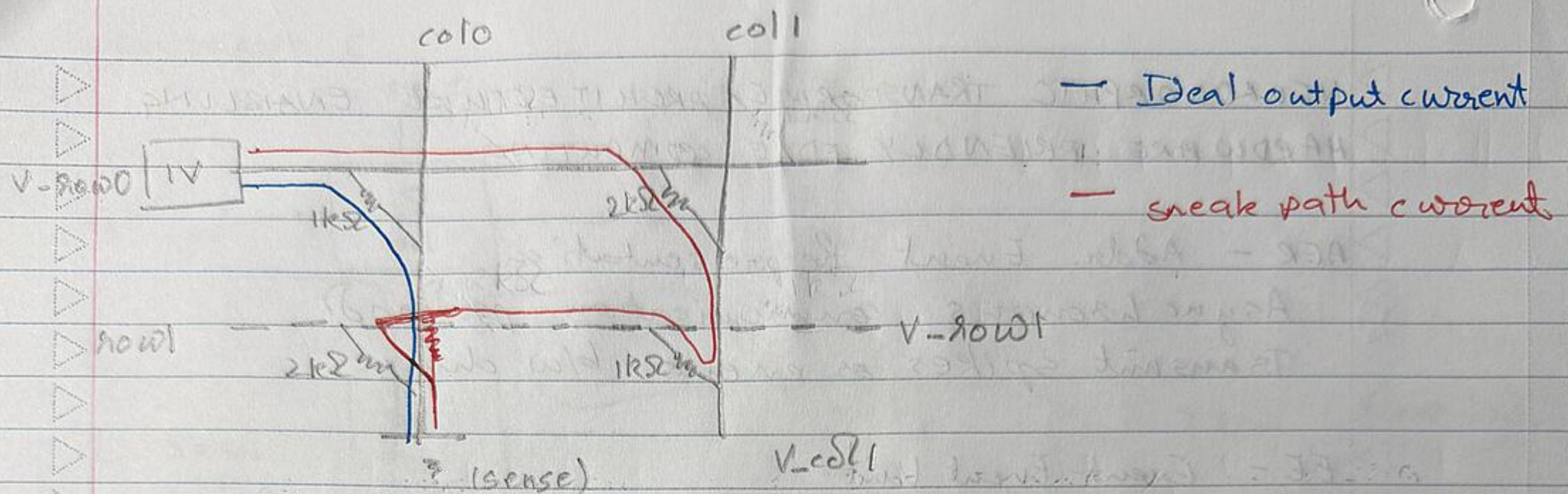


CODEFEST 6



- 1) Given: $V_{row0} = 1V$
 $col0 = 0V$
 $row1 = 0V$
 $col1 = 0V$

Ideal output current (as seen from the diagram) $= \frac{V}{R} = \frac{V_{row0}}{1k\Omega} = 1mA$

$\therefore I_{col0} = 1mA$

- 2) Given $row1$ and $col1 =$ floating (undriven)

Kirchoff's Current Law (KCL) states that total current entering a node (or junction) in an electrical circuit must equal the total current leaving that node.

$$\sum_{k=1}^n I_k = 0$$

KCL at V_{col1} : $\frac{V_{row0} - V_{col1}}{R[0][1]} + \frac{V_{row1} - V_{col1}}{R[1][1]} = 0$

$$\frac{1 - V_{col1}}{2k\Omega} + \frac{V_{row1} - V_{col1}}{1k\Omega} = 0$$

$$1 - V_{col1} + 2V_{row1} - 2V_{col1} = 0$$

$$1 - 3V_{col1} + 2V_{row1} = 0 \quad \text{--- (1)}$$

KCL at V_{row1} : $\frac{V_{row1} - V_{col1}}{R[1\Omega]} + \frac{V_{row1} - V_{col1}}{R[1\Omega]} = 0$

$$\frac{V_{row1} - 0}{2k\Omega} + \frac{V_{row1} - V_{col1}}{1k\Omega} = 0$$

$$V_{row1} + 2V_{row1} - 2V_{col1} = 0$$

$$3V_{row1} - 2V_{col1} = 0$$

$$V_{row1} = \frac{2V_{col1}}{3} \quad \text{--- (2)}$$

Solving from equation (1) and (2)

$$1 - 3V_{col1} + 2\left(\frac{2V_{col1}}{3}\right) = 0$$

$$3 - 9V_{col1} + 4V_{col1} = 0$$

$$5V_{col1} = 3$$

$$V_{col1} = \frac{3}{5} = 0.6 \text{ V}$$

$$V_{row1} = \frac{2}{3} \times V_{col1} = \frac{2}{3} \times \frac{3}{5} = 0.4 \text{ V}$$

$$\text{Sneak path current} = \frac{V}{R_{\text{total}}} = \frac{1V}{R[0][1] + R[1][1] + R[1][0]}$$

$$= \frac{1V}{2k\Omega + 1k\Omega + 2k\Omega} = \frac{1V}{5k\Omega} = 0.2 \text{ mA}$$

$$\begin{aligned} I_{\text{col } 0} &= \text{Ideal output current} + \text{sneak path current} \\ &= 1 \text{ mA} + 0.2 \text{ mA} \\ &= 1.2 \text{ mA} \end{aligned}$$

- 3) In matrix vector multiplication, as the input (voltage) changes on the row) changes, the results are seen in the columns at the bottom of the crossbar. Now, in this case the correct current was 1mA but was recorded 1.2mA due to the sneak path which is 20% error. This crossbar example was a 2x2 matrix with a single sneak path, now imagine the matrix size increases, this would result in more sneak paths and cause an increase in the error.